STA 322: DESIGN AND ANALYSIS OF EXPERIMENT I

MOI UNIVERSITY

DEPARTMENT OF MATHEMATICS, PHYSICS & COMPUTING

COURSE LECTURER: **DR. TUM**
COURSE TITLE AND CODE: **STA 322: Design and Analysis of Experiments I**
Bachelor of Science in Applied Statistics and Arts
YEAR OF STUDY: **3rd YEAR, 2nd SEMESTER**

COURSE OUTLINE CREDIT HOURS: 4 HRS

Purpose of the Course: Is to enable learners acquire knowledge in Design of Experiments.

Course Description

Principles of experimentations: Construction and analysis of common designs in research; completely Randomized Design (CRD), Randomized Complete Block Design (RCBD), Latin square designs. Factorial experiments, BIB. Means separations; Duncan multiple range test (DMRT) and Least significance differences (LSD). Analysis of variance, Missing plot technique

Class Schedule

Session/week	CONTENTS /TOPICS
1	**General Principles of Experimental Designs** Randomization Orthogonal Balance Local Control Sequential Design Estimation of Treatment Contrast and their Precision
2-3	**The Analysis of Variance (ANOVA)** The one way Classification Analysis of Variance Analysis of the fixed effect model Comparison of individual treatment means The random effect model Statistical Analysis of a completely randomized design (CRD) Means separations; Duncan multiple range test (DMRT) and Least significance differences (LSD).
4-5	**The Randomized Complete Block Design (RBD)** Statistical analysis of randomized block design Rationale for Blocking Relative efficiency of randomized block design w.r.t completely randomized design Estimation of missing plots
6	**CAT I-Week 6**
7-8	**Latin Square Design** Rationale ANOVA of latin square design Estimation of missing observation in latin square design Relative efficiency
9	**Factorial Designs** Basic definitions and principles 2^2 factorial design

	2^3 factorial design Calculation & Interpretation of effects and interaction
10	**CAT II -Week 9**
11	**Incomplete Block Design** Introduction Balanced Incomplete block design (BIB) Partially Balanced Incomplete block design (PBIB)
12-13	**EXAMINATIONS Week 12-Week 14**

Teaching Methodology

Lectures, tutorials, assignments, group discussions and presentations

Instructional Materials/Equipment

Audio-visuals devices, computers/internet services, journals, newspapers, chalk/pens and white boards, flip charts, learning centers and markers, projector, laptop.

Student Evaluation Criteria

	Aspects	**Score**
1	Class attendance, participation & assignments	**40%**
2	Continuous Assessment Test(s)	
3	University exam	**60%**
	TOTAL	**100%**

References:

1. John J.A and Quanoulle M.H Experiments Design and Analysis
2. Das, M.N. and Giri, N.C. (1986). Design and Analysis of Experiments, Wiley Eastern.
3. D. C. Montgomery. Design and Analysis of Experiments.John Wiley, NY. 2nd Edition
4. Hinkelman, K. and Kempthorne, A. (1994). Design and Analysis of Experiments. Vol. I, Introduction to experimental design.
5. Statistical Design, G. Casella, Chapman and Hall, 2008)
6. Internet Sources

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COURSE OUTLINE BREAK DOWN

1. **Introduction**
   - ✓ Terminologies
   - ✓ General principles of experimental design
   - ✓ Randomization
   - ✓ Applications
   - ✓ Co-variate
   - ✓ Orthogonal balance
   - ✓ Local control
   - ✓ Sequential design
   - ✓ Estimation of treatment contrast and their precision.
2. **The advantages of variance (ANOVA)**
   - ✓ **Introduction**
   - ✓ The one way classification analysis of variance
   - ✓ Analysis of the fixed effect model
   - ✓ Comparison of individual treatment means
   - ✓ The random effects model
   - ✓ Statistical analysis of a completely randomed design. (C.R.D)
3. **The Randomized Complete Block Design (R.B.D)**
   - ✓ Statistical analysis of randomized Block Design
   - ✓ Rationale for Blocking
   - ✓ Relative efficiency of randomized block design w.r.t completely randomized design (CRD)
   - ✓ Estimation of missing plots.
4. **Latin Square Design**
   - ✓ Rationale ANOVA of Latin square design
   - ✓ Estimation of missing observation I Latin design
   - ✓ Relative efficiency.
5. **Factorial Designs**
   - ✓ Basic definitions
   - ✓ 2^2 factorial design
   - ✓ 2^3 factorial design
   - ✓ Calculations and interpretation of effects and interactions
6. **Complete Block Design**
   - ✓ **Introduction**
   - ✓ Balanced incomplete block design (BIB)
   - ✓ Partially balance incomplete block design (PBIB)

Design and Analysis of Experiment

Introduction

Design of experiment means how to design an experiment in the sense that how the observations or measurements should be obtained to answer a query in a valid, efficient and economical way. The designing of experiment and the analysis of obtained data are inseparable.

If the experiment is designed properly keeping in mind the question, then the data generated is valid and proper analysis of data provides the valid statistical inferences.

If the experiment is not well designed, the validity of the statistical inferences is questionable and may be invalid.

The subject method of design experimental includes the following stages;

i. Planning of experiment
ii. Obtaining relevant information from it regarding the statistical hypothesis under study
iii. Making a statistical analysis of the data
iv. Making meaningful conclusions

Definitions

By the **design of experiment** we refer to the process of planning the experiment so that appropriate data will be collected which may be analyzed by statistical method resulting in valid and objective conclusion.

Terminologies

It is important to understand first the basic terminologies used in the experimental design.

a) **Experiment** is a device of getting an answer to a problem under consideration
   Experiment is an investigation to obtain new information or proving the result of earlier experiment

   i. **Absolute experiment-** consists in determining the absolute value of some characteristics like obtaining average IQ for a group of people or finding correlation coefficient values between two variables.

ii. **Comparative experiment-** is design to compare the effects two or more objects on some population characteristics e.g. compare the yield of different kinds of varieties.

b) **Treatment –** these are various objects of comparison in a comparative experiment e.g. different kinds of varieties of a crop or different types of fertilizer.

Or Procedure that effect a material to be tested and compared with other treatments

Example: Type of fertilizer

- NPK Blue and NPK Yellow

c) **Treatment design:** A treatment design is the manner in which the levels of treatments are arranged in an experiment.

Example: (Ref.: Statistical Design, G. Casella, Chapman and Hall, 2008)

Suppose some varieties of fish food are to be investigated on some species of fishes.

The food is placed in the water tanks containing the fishes.

The response is the increase in the weight of fish.

The experimental unit is the tank, as the treatment is applied to the tank, not to the fish. Note that if the experimenter had taken the fish in hand and placed the food in the mouth of fish, then the fish would have been the experimental unit as long as each of the fish got an independent scoop of food.

d) **Experimental unit –** this is the smallest division of the experimental material to which you can apply the treatment and on which we make observation on the variables under study.

Or this is the unit of material that receives a treatment or where the treatment is given

Example :

- a plant
- an animal
- a square meter plot

e) **Blocks-** are subgroups of an experimental unit which are relatively uniform or homogenous among themselves than the experimental unit as a whole

f) **Yield-** measurement of variable under study on different experimental units

g) **Experimental error-** suppose a large homogenous field is divided into different plots of equal sizes and shapes.

If different treatments are applied to different plots yields may be different and the experimenter would note that the observed differences are due to treatment or by chances or due to uncontrolled factors.

At the same time the yield would vary from one plot to another even if the same treatment is used for the whole plot maybe due to varying soil fertility in different plots such variation which is due to random or chance or non-assigned factors beyond human control is called **experimental error.**

h) **Replication-** is the repetition of the basic experiment or repetition of treatment under investigation.

If treatment is allotted to an experimental unit in an experiment it is said to be replicated r times.

Replication helps in increasing the accuracy of the estimate of the treatment effects.

Or repetition or appearance of a treatment more than once in an experiment.

i) **Variables**-Characteristics of the experimental unit that can be measured

Quantitative or qualitative variables

Discrete or continuous variables

j) **Precision-** is the amount of information in a design and is given by the reciprocal of the variance of the mean.

$$\text{var}(\overline{x}) = \frac{\sigma^2}{n}$$

$$p = \frac{1}{\frac{\sigma^2}{n}} = \frac{n}{\sigma^2}$$

Efficiency of a design- consider two designs D_1 and D_2 with variance σ_1^2 and σ_2^2 and replication r_1 and r_2 respectively. The efficiency of D_1 i.e.

$$E(D_1\, w.r.t\, D_2) = \frac{p_1}{p_2} \qquad p_1 = \frac{r_1}{\sigma_1^2} \qquad p_2 = \frac{r_2}{\sigma_2^2}$$

$$E(D_1 / D_2) = \frac{r_1\sigma_2^2}{r_2\sigma_1^2} = \left(\frac{r_1}{r_2}\right)\frac{\sigma_2^2}{\sigma_1^2}$$

E =1 then **D_1** and **D_2** are equally efficient

E > 1 then $\mathbf{D_2}$ is more efficient than $\mathbf{D_1}$

E < 1 then $\mathbf{D_1}$ is more efficient than $\mathbf{D_2}$

One way analysis of variance ANOVA

T_1	T_3
T_2	

Randomized complete Block design

B_1	B_2	B_3
T_2	T_1	T_2
T_1	T_2	T_3
T_3	T_3	T_1

Latin Square

T_1	T_3	T_2
T_2	T_1	T_3
T_3	T_2	T_1

Factorial design

A – (0,1) 2^2

B – (0,1)

A – (0, 1, 2) 2^3

B – (0, 1, 2)

Design of experiment:

One of the main objectives of designing an experiment is how to verify the hypothesis in an efficient and economical way.

In the contest of the null hypothesis of equality of several means of normal populations having same variances, the analysis of variance technique can be used.

Note that such techniques are based on certain statistical assumptions.

If these assumptions are violated, the outcome of the test of hypothesis then may also be faulty and the analysis of data may be meaningless.

So the main question is how to obtain the data such that the assumptions are met and the data is readily available for the application of tools like analysis of variance.

The designing of such mechanism to obtain such data is achieved by the design of experiment. After obtaining the sufficient experimental unit, the treatments are allocated to the experimental units in a random fashion.

Design of experiment provides a method by which the treatments are placed at random on the experimental units in such a way that the responses are estimated with the utmost precision possible.

Experimental Designs

Arrangement of experimental unit that contains treatments and replications into various designs to estimate and control experimental error so as to interpret results accurately.

The major among experimental designs is the way in which experimental units are classified or grouped.

An experimental design can be simple or complex.

It is, however, advisable to choose a less complicated design that best provides the desired precision.

A pot experiment can be conducted to determine the effect of N rate (0, 45, 90, 135 and 180 kg N ha^{-1}) with four replications on yield of maize cobs

Examples:

Complete Randomized Design (CRD)

Randomized Complete Block Design (RCBD)

Latin Square Design

Split Plot Design

Latin Square Designs +

<u>Selected Latin Squares</u>

3 x 3	4 x 4			
A B C	**A B C D**	**A B C D**	**A B C D**	**A B C D**
B C A	**B A D C**	**B C D A**	**B D A C**	**B A D C**
C A B	**C D B A**	**C D A B**	**C A D B**	**C D A B**
	D C A B	**D A B C**	**D C B A**	**D C B A**

5 x 5	6 x 6
A B C D E	**A B C D E F**
B A E C D	**B F D C A E**
C D A E B	**C D E F B A**
D E B A C	**D A F E C B**
E C D B A	**E C A B F D**
	F E B A D C

Principles of experimental design

There are three basic principles of design which were developed by Sir Ronald A. Fisher.

i. Randomization
ii. Replication
iii. Error control and blocking

Randomization

After the treatment and experimental units are decided the treatment are allotted to the experimental unit at random to avoid any form of subjective (personal) bias in the experiment.

This ensures validity of the results and it also helps to have objective comparisons among treatment.

Randomization therefore ensures that unbiased estimate of treatment effects are obtained.

It also ensures independence of the observations which is necessary for drawing valid inferences from observation by applying appropriate statistical techniques.

The treatments of experimental unit are arrange so that each experimental unit has the same chance to be selected to receive a treatment

In the development of analysis of variance, we assume that the errors are random and independent.

In turn, the observations also become random.

The principle of randomization ensures this.

The random assignment of experimental units to treatments results in the following outcomes.

a) It eliminates the systematic bias.

b) It is needed to obtain a representative sample from the population.

c) It helps in distributing the unknown variation due to confounded variables throughout the experiment and breaks the confounding influence.

Randomization forms a basis of valid experiment but replication is also needed for the validity of the experiment.

If the randomization process is such that every experimental unit has an equal chance of receiving each treatment, it is called a **complete randomization**.

Replication

In the replication principle, any treatment is repeated a number of times to obtain a valid and more reliable estimate than which is possible with one observation only.

Repetition of treatment under investigation, it serves to reduce the experimental error and thus enables us to obtain more précised estimates of the treatment effects.

It also provides an estimate of the error variance or the experimental error without which we cannot test the significance of the difference between any two treatments or determine the length of confidence interval.

The estimate of the experimental error is obtained by considering the differences in the plot receiving the same treatment in different replication.

Replication provides an efficient way of increasing the precision of an experiment.

The precision increases with the increase in the number of observations.

For example, if variance of x is σ^2 then variance of sample mean $\bar{x}$ based on n observation is $\frac{\sigma^2}{n}$ So as n increases, $\text{var}(\bar{x})$ decreases.

Error Control (Local control, Blocking)

This is the process of reducing the experiment error by dividing the relatively heterogeneous experimental area into homogenous blocks.

The replication is used with local control to reduce the experimental error.

Experimental errors will be small if the experimental units are more homogenous.

For example, if the experimental units are divided into different groups such that they are homogeneous within the blocks, than the variation among the blocks is eliminated and ideally the error component will contain the variation due to the treatments only.

This will in turn increase the efficiency.

ANALYSIS OF VARIANCE (ANOVA)

Analysis of variance (ANOVA) is to determine the ratio of between samples to the variance of within samples that is the F distribution.

The value of F is used to reject or accept the null hypothesis.

It is used to analyze the variances of treatments or events for significant differences between treatment variances, particularly in situations where more than two treatments are involved. ANOVA can on only be used to ascertain if the treatment differences are significant or not.

$$F = \frac{s^2, \text{ calculated from sample mean}}{s^2, \text{ calculate from variance between individual sample}}$$

$$= \frac{s_a^2 \text{ (variance between samples)}}{s_d^2 \text{ (variance within samples)}}$$

Null Hypothesis

Statement indicating that a parameter having certain value

Alternative Hypothesis

Statement indicating that a parameter having value that differ from null hypothesis

Critical area

- area to reject null hypothesis

The one way classification of analysis of variance

Suppose we have v treatment to compare. The observed response from each treatment are therefore regarded as a **random variable**.

Let Y_{ij} represent the jth observation under the ith treatment where (j=1,2,...,r) (i=1,2,...,v)

The data appear as follows

Contingency Table

	Observations				
Treatment	**1**	**2**		**r**	**Totals**
1	Y_{11}	Y_{12}	....	Y_{1r}	$Y_{1\bullet}$
2	Y_{21}	Y_{22}	.....	Y_{2r}	$Y_{2\bullet}$
3	Y_{31}	Y_{32}	....	Y_{3r}	$Y_{3\bullet}$
:	:	:		:	:
:	:	:		:	:
v	Y_{v1}	Y_{v2}	....	Y_{vr}	$Y_{v\bullet}$

The total number of observations is Y_{vr}. We can describe the observation using the linear statistical model.

$$Y_{ij} = \mu + \alpha_i + e_{ij}$$

α_i = effect due to i^{th} treatment assume to be a fixed an unknown constant

μ = is the overall or grand mean common to all treatment, it is the mean yield in the absence of the treatment effect and to the experiment

e_{ij} is the random error of the jth observation receiving the ith treatment and is assumed to be normal and independently distributed random variable with mean zero and variance σ^2.

$$e_{ij} \quad N(0,\sigma^2)$$

$$E(Y_{ij}) = E\left[\mu + \alpha_i + e_{ij}\right] = \mu + \alpha_i + E(e_{ij}) = \mu + \alpha_i$$

$$\text{var}(Y_{ij}) = \text{var}(\mu + \alpha_i + e_{ij}) = \text{var}(e_{ij}) = \sigma^2$$

$$Y_{ij} \quad N(\mu + \alpha_i, \sigma^2)$$

In this model where α_i is fixed and the error effect is random then we have a *fixed effect model*. The problem will be to test the appropriate hypothesis about the treatment effect and to estimate them.

Analysis of the fixed effect model

In the fixed effect model α_i's α_i's are usually defined as **deviation from overall mean**

Such that $\sum_{i=1}^{v} \alpha_i = 0$ Since we are interested in testing the equality of the v treatment effects.

The appropriate hypothesis

H_0: $\alpha_i = \alpha_2 = ... = 0$

H_1: $\alpha_i \neq 0$ for any i

This is to say that if the **H_0** is true then

$$Y_{ij} = \mu + e_{ij}$$

To mean each observation is made up of mean μ and the value of e_{ij}

The appropriate test procedure is the ANOVA which will proceed to make for the one way classification fixed model.

$$Y_{ij} = \mu + \alpha_i + e_{ij}$$

We have one way of classification because we have only one factor such an experiment design is called **completely randomized design (CRD)**

<u>Remarks</u>

i. If α_i are random then the model is random effect model
ii. When some α_i are fixed and some are random it is called a **mixed effect model**